Editors' Suggestion

Cavity-Enhanced Solid-State Nuclear Spin Gyroscope

Hanfeng Wang,[1,*] Shuang Wu,[1,2] Kurt Jacobs,[3,4] Yuqin Duan,[1] Dirk R. Englund,[1] and Matthew E. Trusheim[1,3,†]
[1]*Massachusetts Institute of Technology, 50 Vassar Street, Cambridge, Massachusetts 02139, USA*
[2]*Honda Research Institute USA, Inc., San Jose, California 95134, USA*
[3]*U.S. Army DEVCOM Army Research Laboratory, Adelphi, Maryland 20783, USA*
[4]*Department of Physics, University of Massachusetts Boston, Boston, Massachusetts 02125, USA*

(Received 18 December 2024; accepted 7 April 2025; published 9 May 2025)

Solid-state quantum sensors based on ensembles of nitrogen-vacancy (NV) centers in diamond have emerged as powerful tools for precise sensing applications. Nuclear spin sensors are particularly well suited for applications requiring long coherence times, such as inertial sensing, but remain underexplored due to control complexity and limited optical readout efficiency. In this work, we propose cooperative cavity quantum electrodynamic (cQED) coupling to achieve efficient nuclear spin readout. Unlike previous cQED methods used to enhance electron spin readout, here we employ two-field interference in the NV hyperfine subspace to directly probe the nuclear spin transitions. We model the nuclear spin NV-cQED system (nNV-cQED) and observe several distinct regimes, including electromagnetically induced transparency, masing without inversion, and oscillatory behavior. We then evaluate the nNV-cQED system as an inertial sensor, indicating a rotation sensitivity improved by 3 orders of magnitude compared to previous solid-state spin demonstrations. Furthermore, we show that the NV electron spin can be simultaneously used as a comagnetometer, and the four crystallographic axes of NVs can be employed for vector resolution in a single nNV-cQED system. These results showcase the applications of two-field interference using the nNV-cQED platform, providing critical insights into the manipulation and control of quantum states in hybrid NV systems and unlocking new possibilities for high-performance quantum sensing.

DOI: 10.1103/PhysRevLett.134.183603

Introduction—Nitrogen-vacancy (NV) centers in diamond have emerged as a promising platform for a variety of sensing modalities [1–7] due to favorable attributes including room-temperature spin polarization and readout, atomic-scale size [8–10], and long coherence times [11–13]. NV sensors generally use magnetic resonance spectroscopy of the electron ground-state spin triplet transition frequencies. The nuclear spin degree of freedom is less explored as a sensor, since it is not directly accessible for optical polarization and readout, and has potentially disadvantageous properties such as reduced resonant frequency and gyromagnetic ratio. However, reduced sensitivity to noisy magnetic fields results in a long coherence time [14], which is advantageous for quantum memory and inertial sensing applications [15–17].

Nuclear spin gyroscopes promise stable long-term performance with competitive short-term sensitivity. Devices based on atomic vapors have achieved inertial angle random walk (ARW) comparable with the optical sensors at the $\mathrm{mdeg}/\sqrt{\mathrm{h}}$ level, but corresponding performance has not been achieved using nuclei in the solid state [18]. The NV-based solid-state nuclear spin gyroscope describes a standard quantum limit (SQL) as $\eta = 1/\sqrt{NT_2^*}$, where T_2^* is the nuclear spin dephasing time and N is the number of spins [19–21]. This leads to a quantum-limited ARW of $0.1\ \mathrm{mdeg/s}/\sqrt{\mathrm{Hz}}$ for a spin ensemble of $N = 10^{14}$ and $T_2^* = 2$ ms, which outperforms similarly sized (millimeter-scale) solid-state systems such as microelectromechanical systems and fiber optic gyroscopes [22,23]. However, current state-of-the-art NV-based nuclear spin gyroscopes achieve an ARW of $\eta_{\mathrm{exp}} = 4.7\ \mathrm{deg}/\mathrm{s}/\sqrt{\mathrm{Hz}}$ [15,24]. This ARW remains significantly higher than the SQL predicted in the theoretical outline [21] and can be quantified as an inverse readout fidelity $\sigma_n = \eta_{\mathrm{exp}}/\eta \sim 10^4$ that captures all measurement imperfections. This leaves room for dramatic improvement toward the SQL.

NV-cavity quantum electrodynamic (cQED) coupling is a promising avenue that has been applied to enhance readout fidelity for electron spin ensembles [25,26]. However, the manipulation of nuclear spin has not been demonstrated with a cQED scheme, as nuclei are not directly spin polarizable using optics and have low gyromagnetic ratio and resonance frequency that substantially reduces achievable spin-cavity coupling. In this work, we extend our electron spin-based cQED scheme to realize a nuclear spin readout (nNV-cQED) using two-field interference. Like electromagnetically induced transparency (EIT) gyroscopes in atomic vapor systems, this mechanism exhibits a linewidth limited by the nuclear spin coherence,

*Contact author: hanfengw@mit.edu
†Contact author: matthew.e.trusheim.civ@army.mil

 © 2025 American Physical Society

providing an exceptional sensitivity to rotation. We then show that the electron spin coherence of the cavity-coupled transition becomes positive, leading to a cavity gain and enhanced sensitivity in the masing without inversion (MWI) regime [27]. Optimizing over performance and additional spin refrigeration effects [28], we indicate an ARW of 1.5 $\mathrm{mdeg/s/\sqrt{Hz}}$ with an inverse readout fidelity of $\sigma_n \sim 37$, surpassing the sensitivity of state-of-the-art NV gyroscopes by 3 orders of magnitude.

To further enhance functionality, we introduce a "4-EIT" scheme enabling vector inertial sensing and electron spin comagnetometry [24]. The vector resolution is realized via four NV axes, which differs from atomic gyroscopes where external magnetic bias primarily determines nuclei orientation. An electron spin comagnetometer is implemented by measuring common mode and difference-frequency shifts of driving fields, improving bias stability by compensating for noise due to electron spin and temperature. Each transition (rotation and magnetic field) in the 4-EIT scheme can be measured independently with minimal crosstalk or performance degradation. The nNV-cQED system enhances inertial sensitivity while providing a pathway to broader applications exploring quantum phenomena in hybrid solid-state systems.

Two-field interference—The nuclear spin gyroscope relies on two-field interference in a hybrid nNV-cQED system, where a probe microwave field's state is controlled by a driving field via coherent NV spin interaction. Two-field interference phenomena, such as EIT, have been demonstrated for NVs and atomic systems [29–32]. However, EIT in highly cooperative cavity-coupled ensembles is rarely reported, and NV ensembles are further complicated by optical polarization and inhomogeneous broadening.

We consider an NV ensemble in a bulk diamond, as illustrated in Fig. 1(a). The NV ground state comprises an $S = 1$ electronic triplet state coupled via hyperfine interaction to the $I = 1$ ^{14}N nuclear states. Taking the $\{111\}$ as the z axis, the NV Hamiltonian reads [29]

$$\mathcal{H}_{\mathrm{NV}}/h = D\hat{S}_z^2 + \gamma_e \hat{\mathbf{S}}\cdot\mathbf{B} + Q\hat{I}_z^2 - \gamma_n \hat{\mathbf{I}}\cdot\mathbf{B} + \hat{\mathbf{S}}\cdot\underline{\underline{\mathcal{A}}}\cdot\hat{\mathbf{I}}, \quad (1)$$

where $D = 2.87$ GHz and $Q = -4.945$ MHz are the electron and nuclear spin zero-field splittings, respectively, $\gamma_e = 2\pi \times 2.8$ MHz/G and $\gamma_n = 2\pi \times 0.308$ kHz/G are the electron and nuclear spin gyromagnetic ratios, respectively, and $\mathcal{A}$ is the diagonal hyperfine interaction tensor. The NV ground-state level structure based on Eq. (1) is shown in Fig. 1(b). An off-axis magnetic field B mixes the electron and nuclear spin states, enabling forbidden spin-exchange transitions. We focus on a Λ system formed by electron-nuclear spin states $\{|0,-1\rangle, |0,0\rangle, |1,-1\rangle\}$ and label these as $\{|1\rangle, |2\rangle, |\mathrm{e}\rangle\}$, respectively.

We consider the interaction of an ensemble of N NVs with a quantized cavity mode field a coupled to the

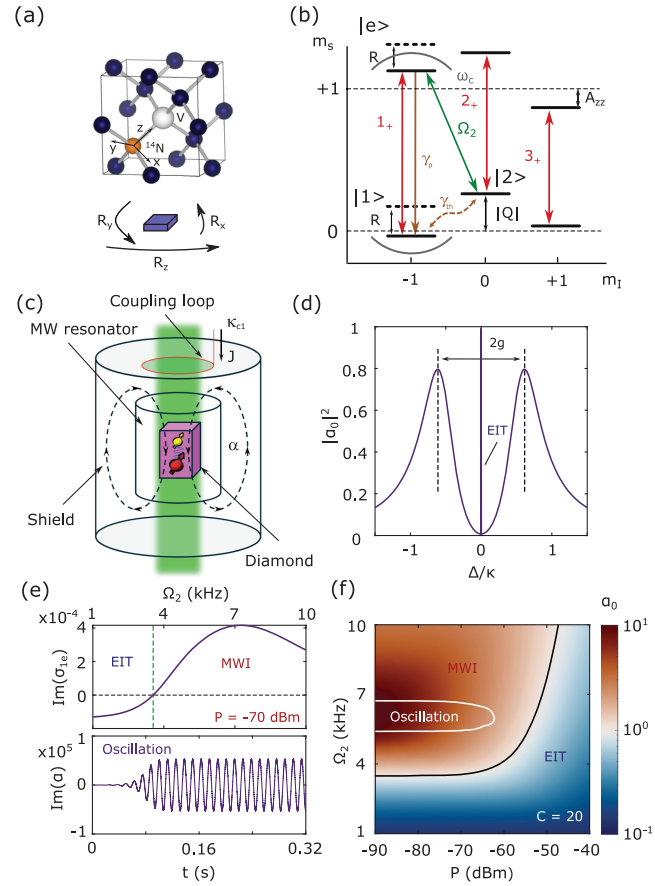

FIG. 1. Two-field interference in nNV-cQED. (a) Top: NV crystal structure. Bottom: diamond with NVs rotates with rate $\mathbf{R} = \{R_x, R_y, R_z\}$. (b) NV energy level structure. The transition $|1\rangle \leftrightarrow |\mathrm{e}\rangle$ is coupled to the cavity mode for the cavity-enhanced readout. A driving field Ω_2 is applied between the spin-exchanging transition $|2\rangle \leftrightarrow |\mathrm{e}\rangle$. (c) Hybrid system with a microwave resonator and an NV spin ensemble. A green laser is applied to continuously polarize the NV spin to the $|m_s = 0\rangle$, and a detection loop is incorporated to measure the reflection signal from the resonator. (d) $|\alpha_0|^2$ as a function of detuning Δ/κ within strong coupling regime. An EIT feature appears around the resonant frequency. (e) Top: $\mathrm{Im}(\sigma_{1\mathrm{e}})$ as a function of Ω_2. The EIT (MWI) regime features a negative (positive) $\mathrm{Im}(\sigma_{1\mathrm{e}})$. Bottom: time dynamics of $\mathrm{Im}(\alpha)$. (f) α_0 as a function of P and Ω_2. The solid line indicates the perfect EIT condition. The boundary of the oscillation regime is marked as a white line.

transition $|1\rangle \leftrightarrow |\mathrm{e}\rangle$, driven by an external probe field J, as shown in Fig. 1(c). The nNV-cQED Hamiltonian reads

$$\mathcal{H} = \Delta a^\dagger a + J(a^\dagger + a) + \sum_{j=1}^{N} \Delta_{\mathrm{s}}^{(j)} \sigma_{\mathrm{ee}}^{(j)} + \Delta_2^{(j)} \sigma_{22}^{(j)} + \sum_{j=1}^{N} g_{\mathrm{s}}^{(j)} (a\sigma_{\mathrm{e1}}^{(j)} + a^\dagger \sigma_{1\mathrm{e}}^{(j)}) + \Omega_2(\sigma_{2\mathrm{e}}^{(j)} + \sigma_{\mathrm{e2}}^{(j)}). \quad (2)$$

In the following analysis, we assume a homogeneous ensemble and, therefore, drop the spin index j. We validate

this approach by comparing it with numerical simulations for an inhomogeneous ensemble [33]. Here, $\Delta = \omega_c - \omega_d$ is the cavity detuning between detection frequency ω_d and the cavity frequency ω_c, $\Delta_s = \omega_s - \omega_d$ is the detuning between spin transition $|1\rangle \leftrightarrow |e\rangle$ frequency ω_s and detection frequency, and $\Delta_2 = \omega_2 + |Q| - \omega_d$ is the detuning between the driving field ω_2 and the probe field ω_d. The rate g_s is the single-spin coupling strength, J is the probe field strength, and Ω_2 is the Rabi frequency of the spin-flip driving field. The total coupling strength of the ensemble is $g = g_s\sqrt{N}$.

We analytically and numerically solve the quantum master equation to obtain the dynamics for nuclear spin manipulation based on the Hamiltonian in Eq. (2), together with loss processes using a Lindbladian open system formalism [25,33]. The cavity mode has a relaxation rate $\kappa = \kappa_c + \kappa_{c1}$, where κ_c is intrinsic and κ_{c1} is coupling to a microwave probe line. The NV ensemble includes several polarization and decoherence processes: optical polarization driving spins to ground states at rate γ_p; electron spin decoherence at rate Γ; thermal depolarization at rate γ_{th}; and nuclear spin dephasing rate γ_{nd}.

The primary parameter of interest for nNV-cQED operation is the cavity field, modulated by the spin ensemble and driving field. We first consider the case where the spin and cavity are on resonance with achievable system cooperativity $C = 4g^2/\kappa\Gamma \sim 20$ [33], and the system is driven with a rate $\Omega_2 = 6$ kHz on resonance with the spin-exchanging transition while probed with power $P = -55$ dBm. The solution for the cavity field as a function of input probe detuning is shown in Fig. 1(d), using an input-normalized cavity field $|\alpha_0|^2 \equiv |\kappa_{c1}\alpha/J|^2$. In this strong coupling regime, the hybridization of the cavity mode with the spin ensemble results in a Rabi splitting of the cavity field maxima, characteristic of the collectively coupled system. A narrow EIT feature appears at the two-photon resonance condition where $\Delta = 0$ due to two-field interference. The EIT linewidth and amplitude are determined by the interplay of driving and probe fields with nuclear spin decoherence. The minimum linewidth is set by the nuclear spin coherent dark state's lifetime, and the EIT contrast compared to the cavity-coupled spin system scales with system cooperativity.

We now consider the nNV-cQED device under different two-field driving conditions. In the steady state, the spin ensemble depends on the cavity field via the relation $\alpha = (J - ig_sN\sigma_{1e})/(\kappa/2)$, where the spin coherence σ_{1e} depends on the spin-exchange driving field Ω_2 as $\sigma_{1e} = (-ig_s\alpha - i\Omega\sigma_{12})/(\Gamma/2)$ (valid when most population is in $|1\rangle$). The relation between Ω_2 and σ_{1e} is shown in Fig. 1(e), top. Low driving fields result in a negative $\mathrm{Im}(\sigma_{1e})$ dominated by the cavity field α, corresponding to imperfect EIT. As the drive Ω_2 increases, ground-state coherence σ_{12} is built corresponding to coherent population trapping of a nuclear spin dark state, and the term $i\Omega\sigma_{12}$ compensates the

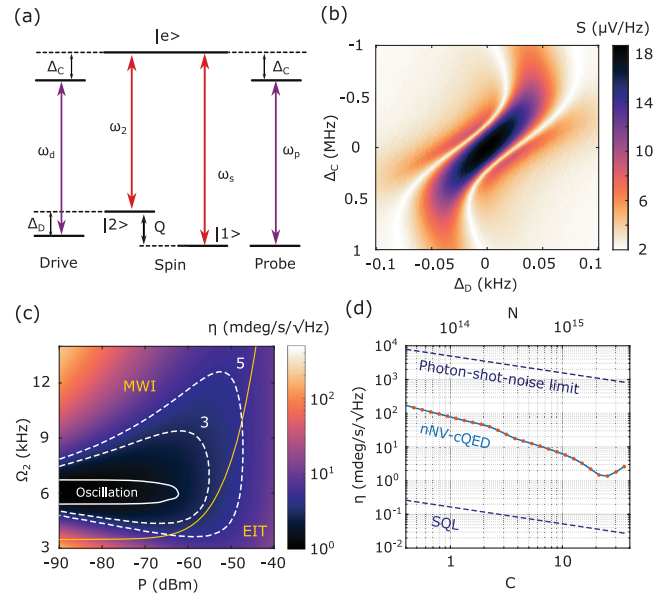

FIG. 2. nNV-cQED gyroscope. (a) Level structure for rotation sensing. The signal can be divided into common-mode signal Δ_C and difference-mode signal Δ_D. (b) S as a function of Δ_C and Δ_D. (c) ARW as a function of probe power P and driving field strength Ω_2. The boundary of the oscillatory regime is marked as a solid white line. An optimal ARW of 1.5 $\mathrm{mdeg/s/\sqrt{Hz}}$ can be achieved. (d) Sensitivity as a function of cooperativity. We plot the photon-shot-noise limit and SQL as a comparison. The best inverse readout fidelity is around $\sigma_n \sim 37$ at $C \sim 20$.

resonant cavity probe field $ig_s\alpha$. This increases σ_{1e} and reduces the phase shift of the cavity field due to spins, with $\sigma_{1e} = 0$ corresponding to complete cancellation (perfect EIT). Interestingly, as Ω_2 continues to increase, σ_{1e} becomes positive, resulting in gain for the cavity field (MWI). This regime produces the largest two-field interference visibilities and is, therefore, the most relevant for sensor operation. In the limit $\Omega_2\sigma_{12} \gg g_s\alpha$, the cavity field has $\alpha \propto (g_sN\Omega_2/\kappa\Gamma) = (\Omega_2/g_s)C$. Increased cooperativity then results in increased MWI gain for fixed driving field.

These regimes are illustrated in Fig. 1(f), which shows the normalized steady-state cavity field α_0 under varying two-field drive parameters. As probe power increases, the system demonstrates saturation effects. In the low-power regime, increased probe power does not destroy the perfect EIT condition, as coherence σ_{12} compensates increased $g_s\alpha$. As $P \to -55$ dBm and beyond, however, increased Ω_2 is required to maintain perfect EIT due to σ_{12} saturation. This combination induces Autler-Townes splitting in the dressed state picture, broadening the resonant peak and decreasing EIT contrast [35].

In this high-cooperativity device, an oscillatory regime exists where $ig_sN\sigma_{1e}$ exceeds the probe field J, providing sustaining gain that overcomes cavity losses. Compared with the EIT and MWI regimes, where output frequency locks with the probe field, frequency in the oscillating

regime is determined by spin-cavity parameters and the driving field. In Fig. 1(e), bottom, we plot $\mathrm{Im}(\alpha)$ as a function of time for detuning $\Delta_b = 2\pi \times 80$ Hz in the oscillating regime, showing a beat note with frequency Δ_b compared to the probe field. We note that sustained oscillation without inversion has rarely been reported in the literature [36,37].

Gyroscope performance—Having shown the features of the nNV-cQED system under two-field drive, we now consider performance in an inertial sensing application. The nuclear spin behavior under a rotation $\mathbf{R}$, shown in Figs. 1(a) and 1(b), is described by the Hamiltonian $\mathcal{H}_R = \mathbf{R} \cdot \mathbf{I}$ [24]. We divide the sensing signal into common mode (Δ_C) and difference mode (Δ_D), as shown in Fig. 2(a). The common mode describes the sensing signal shifting the state $|e\rangle$ while keeping nuclear spin energies constant. The difference mode describes the sensing signal shifting $|2\rangle$ while changing nuclear energy difference from $|Q|$ to $|Q| + \Delta_\mathrm{D}$. A rotation R results in both common- and difference-mode shifts equal to the rotation rate, but the resonant linewidth for the difference mode is limited by nuclear spin linewidth (~ 80 Hz), 3 orders smaller than the electron spin linewidth-limited common signal ($\sim$0.33 MHz).

The device's sensitivity is determined by the signal and noise when probing the system: $\eta = \mathcal{L}/\gamma_e S$. The signal per unit frequency shift is defined as $S = [V\partial\,\mathrm{Im}(r)]/\partial\Delta_\mathrm{D}$, where $V = J\sqrt{\hbar\omega_d R/\kappa_{c1}}$ is the probe field input voltage (R is the resistance) and $r = -1 + \alpha_0$ is the probe reflection coefficient. We plot S as a function of Δ_C and Δ_D in Fig. 2(b) with the same spin-cavity parameters as in the previous section. The optimal S is 18 μV/Hz with a linear dynamical range of 20 Hz.

The noise performance is typically limited by Johnson-Nyquist noise at room temperature $\mathcal{L}_\mathrm{JN}$. However, as shown in previous works [25,28,38,39], the NV ensemble can act as a spin refrigerator, performing below the Johnson-Nyquist limit even with active microwave probes. The noise performance can then be expressed as $\mathcal{L} = \xi\mathcal{L}_\mathrm{JN}$, with the refrigeration prefactor $\xi = 0.45$ [33].

We plot the sensitivity η as a function of Ω_2 and P in Fig. 2(c). An optimal sensitivity of 1.5 mdeg/s/$\sqrt{\mathrm{Hz}}$ is achievable in the MWI regime, while in the EIT regime the best sensitivity is 3.5 mdeg/s/$\sqrt{\mathrm{Hz}}$. In the oscillation regime, the field frequency is determined by spin-cavity parameters and the driving field. The sensitivity of such a frequency modulation system depends on oscillator phase noise [40], which is outside the scope of our discussion. The rotation SQL of this system is 0.04 mdeg/s/$\sqrt{\mathrm{Hz}}$, and the optimum sensitivity corresponds to an inverse readout fidelity of $\sigma_n = 37$, 3 orders better than photon-shot-noise-limited sensitivity [15].

Finally, we analyze the ARW dependence on nNV-cQED system cooperativity. In Fig. 2(d), we plot the ARW dependence with C, given the same single-spin

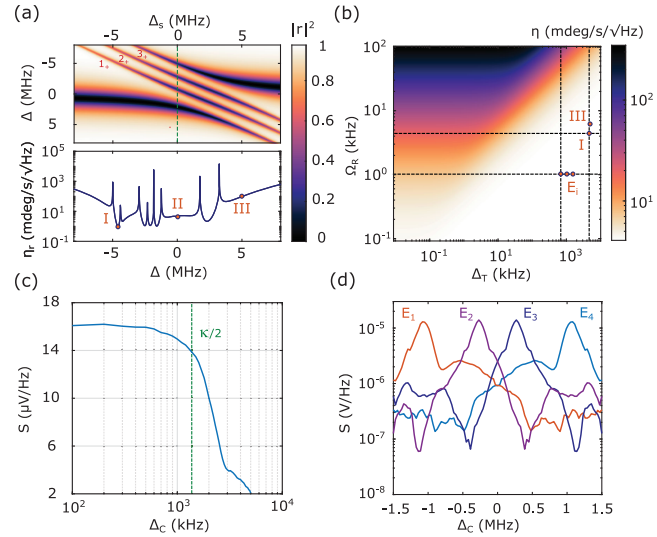

FIG. 3. Comagnetometry and vector gyroscope. (a) Electron spin-based comagnetometer. Top: the reflection $|r|^2$ as a function of Δ and Δ_s. Bottom: the line cut for the sensitivity along $\Delta = 0$ for $P_2 = -40$ dBm. The numerals indicate the frequencies of (I) comagnetometer probe field, (II) EIT probe field and (III) EIT driving field. (b) η in the presence of an additional field with Rabi rate Ω_R and detuning Δ_T. The red dots indicate field combinations for comagnetometry and vector operation. (c) Signal S as a function of Δ_C. S drops significantly beyond the cavity linewidth. (d) The response S for subensembles E_i as a function of Δ_C.

coupling strength and loaded quality factor. Here, cooperativity is tuned by changing the ensemble size. In the low-C regime ($C < 10$), the ARW improves linearly with C. The cooperativity is then decreased superlinearly in the regime of $10 < C < 24$, attributed to MWI. After $C > 24$, the maximum-sensitivity operating point enters the oscillating regime and degrades performance. The best readout fidelity for our system is $\sigma_n = 37$ with $C = 20$ and ensemble size 2.4×10^{15}. Such ensemble size and cooperativity are achievable based on prior experimental works [26,41]. The sensitivity is independent of the g_s^2/κ ratio for fixed cooperativity, and further sensitivity dependence on spin-cavity parameters is detailed in Supplemental Material [33].

Comagnetometry—One of the main challenges for spin-based gyroscopes in ambient conditions is the magnetic field environment. The magnetic field generates a varying nuclear phase via $\gamma_n B_z$, indistinguishable from the rotation signal in the Hamiltonian, therefore setting the noise floor η_r. The concept of a comagnetometer involves utilizing the electron spin to measure the magnetic field, which is then used to compensate for nuclear spin phase accumulation induced by the magnetic environment [24].

Here, we apply a second probe tone to measure the electron spin frequency similarly to NV-cQED work [25]. Figure 3(a) shows cavity reflection $|r|^2$ as a function of Δ and Δ_s. Three avoid-crossing features are observed, corresponding to the three ^{14}N hyperfine transitions, assuming

each subensemble has an equal population with $C = 10$. We plot the magnetic sensitivity-limited ARW floor produced by the second probe tone as a function of its detuning Δ in Fig. 3(a), bottom. With the two-field interference applied in the $|1\rangle$, $|2\rangle$ subspace, however, the coupling strengths become imbalanced. The driving field polarizes the nuclear spin transition to state $|1\rangle$ from $|2\rangle$, reducing the coupling strength for the 2_+ transition compared to 1_+ and producing an asymmetry in the sensitivity spectrum. The comagnetometer sensitivity is maximized at a probe detuning of $\Delta = -4.7$ MHz (point I), where the noise floor scales with comagnetometer probe power as $\eta_r = 1\ \mathrm{mdeg/s}/\sqrt{\mathrm{Hz}}$ with power $P_2 = -40$ dBm. This resonates with the 3_+ transition, effectively using the third nuclear spin hyperfine subensemble as a comagnetometer.

The sensitivity degradation induced by the second probe, detuned from the EIT probe by a frequency detuning Δ_T and Rabi frequency Ω_R, is plotted in Fig. 3(b) [33]. The ideal comagnetometry operating point is shown by the red dot I, with a Rabi frequency $\Omega_R = 4.4$ kHz producing a $\eta_r = 1\ \mathrm{mdeg/s}/\sqrt{\mathrm{Hz}}$. At this point, the EIT rotational sensitivity is not affected by the comagnetometer probe, with $\Delta\eta/\eta < 1\%$. The compensation for the magnetic field drift also provides potential for long-term stability for the nNV-cQED system.

Vector gyroscope—We extend the nNV-cQED system to a vector gyroscope with a multifield EIT scheme. The external magnetic bias is adjusted such that the electron spin transitions are near the cavity resonance yet resolvable ($E_i - E_j > \Gamma$), with multiple fields applied for each spin subensemble. Each subensemble, with the same cooperativity as in the previous section, measures the rotation projection on its axis, as the external bias is small compared to the electron zero-field splitting. Subensembles must be densely packed within the cavity linewidth, since the rotation signal S drops significantly beyond, as shown in Fig. 3(c).

Two kinds of crosstalk may affect the nNV-cQED vector device. First, since the EIT condition is simultaneously met for all subensembles detuned in common-mode frequency, we need to ensure that the signal measured from a particular EIT probe can be associated with shifts of one ensemble. To this end, we consider the signal response S from an EIT probe at frequency Δ_C associated with an energy shift of each subensemble, given that the ensembles are separated by 0.8 MHz. As shown in Fig. 3(d), an EIT probe on resonance with the central ensemble under these conditions would have a maximum relative sensitivity of 10% to the nearest off-resonant ensembles (typically, 0.5%). A linear crosstalk elimination process can be performed for further correction.

Second, the comagnetometer and multiple closely packed EIT probes will induce additional interference effects that degrade performance, as discussed in the comagnetometer session. The sensitivity degradation is also within 1% indicated in Fig. 3(b) (dots labeled E_i).

Conclusion and discussion—Our proposed nNV-cQED gyroscope scheme shows that nuclear spin readout can performed using a strongly coupled microwave cavity, with readout fidelity of $\sigma_n = 37$ outperforming current ensemble optical approaches. The extension of nNV-cQED to multiple ensembles enables comagnetometry and vector sensing, which is not currently achieved in atomic inertial sensors or NV electron-spin cQED experiments. Finally, we show that the nNV-cQED system operating with high cooperativity enters the counterintuitive MWI regime, potentially into an oscillatory phase.

Future work will explore extending the two-field theory to inhomogeneous ensembles, especially in nonlinear, bistable, and MWI regimes; optimizing spin refrigeration and system parameters to approach the quantum limit; and alternative modalities, including applications in microwave memories and timekeeping [42,43].

Acknowledgments—The authors thank Bo-Han Wu for fruitful discussions. H. W. acknowledges support from Robert Bosch GmbH. S. W. acknowledges support from Honda Research Institute USA. Y. D. acknowledges support from Mathworks Fellowship. D. R. E. acknowledges funding from the MITRE Corporation and the United States National Science Foundation Center for Ultracold Atoms.

Data availability—The data that support the findings of this Letter are openly available [44].

[1] J. F. Barry, J. M. Schloss, E. Bauch, M. J. Turner, C. A. Hart, L. M. Pham, and R. L. Walsworth, Sensitivity optimization for NV-diamond magnetometry, Rev. Mod. Phys. **92**, 015004 (2020).

[2] C. L. Degen, F. Reinhard, and P. Cappellaro, Quantum sensing, Rev. Mod. Phys. **89**, 035002 (2017).

[3] M. W. Doherty, N. B. Manson, P. Delaney, F. Jelezko, J. Wrachtrup, and L. C. Hollenberg, The nitrogen-vacancy colour centre in diamond, Phys. Rep. **528**, 1 (2013).

[4] L. Kim, H. Choi, M. E. Trusheim, H. Wang, and D. R. Englund, Nanophotonic quantum sensing with engineered spin-optic coupling, Nanophotonics **12**, 441 (2023).

[5] H. Wang, M. E. Trusheim, L. Kim, H. Raniwala, and D. R. Englund, Field programmable spin arrays for scalable quantum repeaters, Nat. Commun. **14**, 704 (2023).

[6] V. M. Acosta, E. Bauch, M. P. Ledbetter, C. Santori, K.-M. Fu, P. E. Barclay, R. G. Beausoleil, H. Linget, J. F. Roch, F. Treussart *et al.*, Diamonds with a high density of nitrogen-vacancy centers for magnetometry applications, Phys. Rev. B **80**, 115202 (2009).

[7] J. Rovny, S. Gopalakrishnan, A. C. B. Jayich, P. Maletinsky, E. Demler, and N. P. de Leon, Nanoscale diamond quantum sensors for many-body physics, Nat. Rev. Phys. **6**, 753 (2024).

[8] C. Du, T. Van der Sar, T. X. Zhou, P. Upadhyaya, F. Casola, H. Zhang, M. C. Onbasli, C. A. Ross, R. L. Walsworth, Y. Tserkovnyak *et al.*, Control and local measurement of the spin chemical potential in a magnetic insulator, Science **357**, 195 (2017).

[9] M. Pelliccione, A. Jenkins, P. Ovartchaiyapong, C. Reetz, E. Emmanouilidou, N. Ni, and A. C. Bleszynski Jayich, Scanned probe imaging of nanoscale magnetism at cryogenic temperatures with a single-spin quantum sensor, Nat. Nanotechnol. **11**, 700 (2016).

[10] Y. Hu, S. Nagle, Y. Duan, H. Wang, and D. Englund, Developing 3D models of atom-like defect spin memories in crystals for quantum technology research and education, J. Chem. Educ. **101**, 3522 (2024).

[11] E. Bauch, C. A. Hart, J. M. Schloss, M. J. Turner, J. F. Barry, P. Kehayias, S. Singh, and R. L. Walsworth, Ultralong dephasing times in solid-state spin ensembles via quantum control, Phys. Rev. X **8**, 031025 (2018).

[12] P. L. Stanwix, L. M. Pham, J. R. Maze, D. Le Sage, T. K. Yeung, P. Cappellaro, P. R. Hemmer, A. Yacoby, M. D. Lukin, and R. L. Walsworth, Coherence of nitrogen-vacancy electronic spin ensembles in diamond, Phys. Rev. B **82**, 201201(R) (2010).

[13] G. Wang, C. Li, and P. Cappellaro, Observation of symmetry-protected selection rules in periodically driven quantum systems, Phys. Rev. Lett. **127**, 140604 (2021).

[14] N. Kalb, P. C. Humphreys, J. J. Slim, and R. Hanson, Dephasing mechanisms of diamond-based nuclear-spin memories for quantum networks, Phys. Rev. A **97**, 062330 (2018).

[15] A. Jarmola, S. Lourette, V. M. Acosta, A. G. Birdwell, P. Blümler, D. Budker, T. Ivanov, and V. S. Malinovsky, Demonstration of diamond nuclear spin gyroscope, Sci. Adv. **7**, eabl3840 (2021).

[16] G. Wang, M.-T. Nguyen, and P. Cappellaro, Hyperfine-enhanced gyroscope based on solid-state spins, Phys. Rev. Lett. **133**, 150801 (2024).

[17] G. Wang, M.-T. Nguyen, D. W. deQuilettes, E. Price, Z. Hu, D. A. Braje, and P. Cappellaro, Emulated nuclear spin gyroscope with 15 N-V centers in diamond, Phys. Rev. Appl. **22**, 044016 (2024).

[18] J. K. Stockton, K. Takase, and M. A. Kasevich, Absolute geodetic rotation measurement using atom interferometry, Phys. Rev. Lett. **107**, 133001 (2011).

[19] M. P. Ledbetter, K. Jensen, R. Fischer, A. Jarmola, and D. Budker, Gyroscopes based on nitrogen-vacancy centers in diamond, Phys. Rev. A **86**, 052116 (2012).

[20] D. Maclaurin, M. W. Doherty, L. C. L. Hollenberg, and A. M. Martin, Measurable quantum geometric phase from a rotating single spin, Phys. Rev. Lett. **108**, 240403 (2012).

[21] A. Ajoy and P. Cappellaro, Stable three-axis nuclear-spin gyroscope in diamond, Phys. Rev. A **86**, 062104 (2012).

[22] Y.-H. Lai, M.-G. Suh, Y.-K. Lu, B. Shen, Q.-F. Yang, H. Wang, J. Li, S. H. Lee, K. Y. Yang, and K. Vahala, Earth rotation measured by a chip-scale ring laser gyroscope, Nat. Photonics **14**, 345 (2020).

[23] M. P. Hokmabadi, A. Schumer, D. N. Christodoulides, and M. Khajavikhan, Non-Hermitian ring laser gyroscopes with enhanced sagnac sensitivity, Nature (London) **576**, 70 (2019).

[24] V. V. Soshenko, S. V. Bolshedvorskii, O. Rubinas, V. N. Sorokin, A. N. Smolyaninov, V. V. Vorobyov, and A. V. Akimov, Nuclear spin gyroscope based on the nitrogen vacancy center in diamond, Phys. Rev. Lett. **126**, 197702 (2021).

[25] H. Wang, K. L. Tiwari, K. Jacobs, M. Judy, X. Zhang, D. R. Englund, and M. E. Trusheim, A spin-refrigerated cavity quantum electrodynamic sensor, Nat. Commun. **15**, 10320 (2024).

[26] E. R. Eisenach, J. F. Barry, M. F. O'Keeffe, J. M. Schloss, M. H. Steinecker, D. R. Englund, and D. A. Braje, Cavity-enhanced microwave readout of a solid-state spin sensor, Nat. Commun. **12**, 1357 (2021).

[27] Y. Zhu, Lasing without inversion in a closed three-level system, Phys. Rev. A **45**, R6149 (1992).

[28] D. P. Fahey, K. Jacobs, M. J. Turner, H. Choi, J. E. Hoffman, D. Englund, and M. E. Trusheim, Steady-state microwave mode cooling with a diamond N-V ensemble, Phys. Rev. Appl. **20**, 014033 (2023).

[29] P. Huillery, J. Leibold, T. Delord, L. Nicolas, J. Achard, A. Tallaire, and G. Hétet, Coherent microwave control of a nuclear spin ensemble at room temperature, Phys. Rev. B **103**, L140102 (2021).

[30] P. Jamonneau, G. Hétet, A. Dréau, J.-F. Roch, and V. Jacques, Coherent population trapping of a single nuclear spin under ambient conditions, Phys. Rev. Lett. **116**, 043603 (2016).

[31] J. Vanier, Atomic clocks based on coherent population trapping: A review, Appl. Phys. B **81**, 421 (2005).

[32] H. Gray, R. Whitley, and C. Stroud, Coherent trapping of atomic populations, Opt. Lett. **3**, 218 (1978).

[33] See Supplemental Material at http://link.aps.org/supplemental/10.1103/PhysRevLett.134.183603 for additional details for analytical deviations for the nuclear spin state mixing, two-field interference, and sensitivity optimization, which includes Ref. [34].

[34] M. O. Scully and M. S. Zubairy, *Quantum Optics* (Cambridge University Press, Cambridge, England, 1997).

[35] C. N. Cohen-Tannoudji, The autler-townes effect revisited, in *Amazing Light: A Volume Dedicated to Charles Hard Townes on His 80th Birthday* (Springer, New York, 1996), pp. 109–123.

[36] A. S. Zibrov, M. D. Lukin, D. E. Nikonov, L. Hollberg, M. O. Scully, V. Velichansky, and H. G. Robinson, Experimental demonstration of laser oscillation without population inversion via quantum interference in Rb, Phys. Rev. Lett. **75**, 1499 (1995).

[37] J. Mompart and R. Corbalan, Lasing without inversion, J. Opt. B **2**, R7 (2000).

[38] T. Day, M. Isarov, W. J. Pappas, B. C. Johnson, H. Abe, T. Ohshima, D. R. McCamey, A. Laucht, and J. J. Pla, Room-temperature solid-state maser amplifier, Phys. Rev. X **14**, 041066 (2024).

[39] W. Ng, X. Xu, M. Attwood, H. Wu, Z. Meng, X. Chen, and M. Oxborrow, Move aside pentacene: Diazapentacene-doped para-terphenyl, a zero-field room-temperature maser with strong coupling for cavity quantum electrodynamics, Adv. Mater. **35**, 2300441 (2023).

[40] J. F. Barry, R. A. Irion, M. H. Steinecker, D. K. Freeman, J. J. Kedziora, R. G. Wilcox, and D. A. Braje, Ferrimagnetic oscillator magnetometer, Phys. Rev. Appl. **19**, 044044 (2023).

[41] A. Angerer, S. Putz, D. O. Krimer, T. Astner, M. Zens, R. Glattauer, K. Streltsov, W. J. Munro, K. Nemoto, S. Rotter *et al.*, Ultralong relaxation times in bistable hybrid quantum systems, Sci. Adv. **3**, e1701626 (2017).

[42] Y.-C. Wei, B.-H. Wu, Y.-F. Hsiao, P.-J. Tsai, and Y.-C. Chen, Broadband coherent optical memory based on electromagnetically induced transparency, Phys. Rev. A **102**, 063720 (2020).

[43] M. E. Trusheim, K. Jacobs, J. E. Hoffman, D. P. Fahey, D. A. Braje, and D. R. Englund, A polariton-stabilized spin clock, arXiv:2009.02427.

[44] H. Wang, S. Wu, K. Jacobs, Y. Duan, D. R. Englund, and M. E. Trusheim, Data associated with the cavity-enhanced solid-state nuclear spin gyroscope, https://github.com/hanfengw/nNVcQED (2025).